

The Great Story & The Jam Sandwich Robot

SMART START MONTESSORI: COMPUTER SCIENCE LESSON CARD

CORE CONCEPT

Human Robots & Sequential Logic

MONTESSORI CONNECTION

Practical Life (Grace & Courtesy, Sequence of Work)

SETUP & ENVIRONMENT PREP

Set up a clean demonstration table at the front of the circle rug. Unpack bread, a jar of jam with the lid tightly closed, a butter knife, and plates. Prepare paper clipboards for Upper Elementary coders.

KEY MATERIALS

1 loaf of sliced bread
1 jar of jam (sealed)
1 butter knife & paper plates
Paper & pencils (Elementary)
Command Cards (Primary)

CS UNPLUGGED ACTIVITY

The Jam Sandwich Robot

- 01 Gather students on the circle rug and tell the Great Story of how humans communicate with computers.
- 02 Introduce the 'Teacher Robot' who only understands literal, step-by-step commands.
- 03 Place bread, butter, and jam on the table. Ask Primary students to give one physical command at a time.
- 04 Have Upper El students write down a 5-step sandwich algorithm before testing it on the robot.
- 05 Observe as the Robot follows instructions literally (e.g., placing unopened jam jar on top of bread if 'put jam on bread' is commanded).

PRIMARY ADAPTATION

Ages 3–6

1-step physical actions; physically perform step before giving command to Teacher Robot.

ELEMENTARY ADAPTATION

Ages 6–12

Pair work writing precise written algorithm on paper prior to live testing on Teacher Robot.

CONTROL OF ERROR / SELF-CORRECTION

Literal execution by the Teacher Robot. If an instruction is missing 'open the lid', the robot attempts to spread the sealed jar, making the bug immediately obvious without teacher correction.

Kidbots Part 1: The Floor Grid & Arrows

SMART START MONTESSORI: COMPUTER SCIENCE LESSON CARD

CORE CONCEPT

Directional Cards & Precise Instructions

MONTESSORI CONNECTION

Spatial Awareness & Sensorial Orientation

SETUP & ENVIRONMENT PREP

Tape a 3x3 (Primary) and 5x5 (Elementary) grid on the floor using painter's masking tape. Place target objects (plush toy or beanbag) in destination squares. Lay out directional cards.

KEY MATERIALS

Painter's masking tape, Arrow cards (Forward, Turn Left, Turn Right), Destination target objects (beanbags/toys), Command layout mats

CS UNPLUGGED ACTIVITY

Kidbots Floor Grid & Arrow Navigation

- 01 Set up a masking-tape floor grid (3x3 for Primary, 5x5 for Elementary) and place a target object in one cell.
- 02 Designate one student as the 'Robot' and another as the 'Programmer'.
- 03 For Primary: The Programmer walks alongside the Robot, placing physical Arrow Cards (Forward, Turn Left, Turn Right) square by square.
- 04 For Elementary: The Programmer creates a sequence on an external Command Mat before sending the Robot on its path.
- 05 Review precision: Ensure turns (90 degrees) happen in-place without moving forward.

PRIMARY ADAPTATION

Ages 3–6

3x3 Floor Grid. Concrete action: place physical arrow cards directly on grid squares ahead of Robot child.

ELEMENTARY ADAPTATION

Ages 6–12

5x5 Floor Grid. Abstract action: programmer lays out full arrow card sequence on external mat before Robot moves.

CONTROL OF ERROR / SELF-CORRECTION

The physical grid lines. If the Robot steps off the grid or lands in a blank cell away from the target, the Programmer can re-trace their arrow sequence.

Kidbots Part 2: Obstacles & Debugging

SMART START MONTESSORI: COMPUTER SCIENCE LESSON CARD

CORE CONCEPT

Systematic Error Spotting & Debugging

MONTESSORI CONNECTION

Control of Error & Self-Correction

SETUP & ENVIRONMENT PREP

Maintain floor grids from W2. Place 1 cone on the 3x3 grid and 3 cones on the 5x5 grid. Prepare intentional buggy algorithm cards to test student debugging skills.

KEY MATERIALS

Traffic cones or red felt obstacle pads
Floor grid tape
Arrow card sets
Debugging logs & clipboards

CS UNPLUGGED ACTIVITY

Kidbots Obstacles & Debugging

- 01 Introduce obstacle cones or red felt pads on the grid map.
- 02 Intentionally present a faulty arrow card sequence that leads the Robot straight into an obstacle.
- 03 Guide students to 'pause' the robot at the exact step where the error occurs.
- 04 Have students isolate the bad card, replace it with the correct directional card, and resume execution.
- 05 Challenge Upper EI students to find the shortest possible algorithm (fewest total steps).

PRIMARY ADAPTATION

Ages 3–6

3x3 Grid with 1 obstacle cone.
Physically swap out single incorrect arrow card when robot hits obstacle.

ELEMENTARY ADAPTATION

Ages 6–12

5x5 Grid with 3 obstacles. Identify bugs in peer sequences and compare two valid paths to determine efficiency.

CONTROL OF ERROR / SELF-CORRECTION

Physical obstacle cones. Collision with a cone halts the Robot immediately, indicating an error in the current card position.

Rescue Mission: Loops, Functions & Algorithmic Efficiency

SMART START MONTESSORI: COMPUTER SCIENCE LESSON CARD

CORE CONCEPT

Repeat Commands & Subroutines/Function Blocks

MONTESSORI CONNECTION

Sensorial Patterns & Math Skip Counting

SETUP & ENVIRONMENT PREP

Lay out an extended linear grid track on the rug with repeating path sections. Place 'Rescue Target' at the end. Prepare 'Repeat (N)' loop cards and Function mats.

KEY MATERIALS

Extended tape grid tracks
Repeat loop card blocks (2x, 3x, 4x)
Function definition cards
Dry-erase markers & boards

CS UNPLUGGED ACTIVITY

Rescue Mission: Loops & Subroutines

- 01 Set up a longer grid course with repeating path patterns (e.g., 3 forward steps, or repeating stair patterns).
- 02 Introduce the Repeat Loop card and Function definition mats.
- 03 Demonstrate how 3 'Forward' cards can be represented by a single '3x Forward' loop block.
- 04 Have Elementary students write a Function block on a dry-erase card, then call that function name inside their main algorithm.
- 05 Run the Robot child through the compressed program.

PRIMARY ADAPTATION

Ages 3–6

Physical rhythm: clap or stomp for repeated actions ('Step, Clap, Step, Clap' -> 'Repeat 2 times').

ELEMENTARY ADAPTATION

Ages 6–12

Re-write 10-step Kidbot path using Repeat Commands and Subroutines/Function Blocks to condense code under 5 total blocks.

CONTROL OF ERROR / SELF-CORRECTION

Step counting. If a loop executes 3 times instead of 2, the Robot overshoots the target square on the grid.